# Practice workbook · Explain how respiration releases energy and ATP transfers it

Why cells respire, measuring respiration rate, and the ADP/ATP cycle that passes energy to activities like contraction.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-RESPIRATION-ENERGY-ATP

# 1. Question 7 · part b · 5 marks

June 2024 Paper 2R

Worked-example retry · Attempt part b only.

**7** (a) Aerobic and anaerobic respiration both take place during exercise.

(i) Give the word equation for anaerobic respiration.

**(1)**

.....

(ii) Complete and balance the chemical symbol equation for aerobic respiration.

**(2)**

$C_6H_{12}O_6$ + ..... ⟶ ..... + ..... $CO_2$

(iii) A student investigates the rate of carbon dioxide production during exercise and at rest.

This is the equipment they use.

- four pieces of glass tubing
- two boiling tubes
- two rubber bungs
- mouthpiece
- limewater

Draw a diagram to show how the student should set up the equipment to carry out the investigation.

**(2)**

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**20**

# 1. Question 7 · part b · 5 marks

June 2024 Paper 2R

Worked-example retry · Attempt part b only.

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(iv) Describe how a student could use this equipment and a timer to compare the rate of carbon dioxide production during exercise and at rest.

**(4)**

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# 1. Question 7 · part b · 5 marks

June 2024 Paper 2R

Worked-example retry · Attempt part b only.

(b) Aerobic and anaerobic respiration both produce ATP.

Describe the role of ATP in energy transfer during exercise.

**(5)**

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**(Total for Question 7 = 14 marks)**

## 1. Question 7 · part b · 5 marks · Working and answer

Working and answer

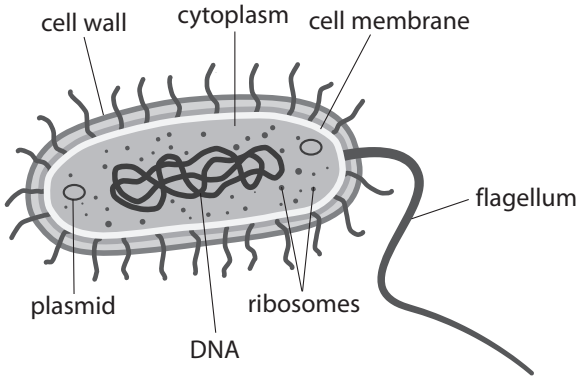
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## 2. Question 3 · part b.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.iii only.

**3** (a) The diagram shows a type of bacterial cell.




(i) The boxes give some structures found in the bacterial cell and some functions of cell structures.

Draw one straight line from each structure to its correct function.

**(3)**

| Structure | Function |
|---|---|
| cell membrane | protects against pathogens |
| | helps the cell to move |
| ribosomes | makes proteins |
| | holds the genetic code |
| flagellum | controls which substances enter and exit the cell |

(ii) State two structural differences between the bacterial cell and a sperm cell.

**(2)**

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2 .....

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**8**

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## 2. Question 3 · part b.iii · 2 marks

January 2020 Paper 1

Further practice · Attempt part b.iii only.

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(b) Many bacteria absorb oxygen and nutrients from their environment. Oxygen and nutrients are used by bacteria in aerobic respiration.

(i) Which process transports oxygen into bacterial cells?

(1)

- ☒ **A** active transport
- ☒ **B** diffusion
- ☒ **C** excretion
- ☒ **D** osmosis

(ii) Complete the word equation for aerobic respiration.

(2)

oxygen + ..... → ..... + water

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(iii) Explain why respiration is important for bacteria.

(2)

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**(Total for Question 3 = 10 marks)**

## 2. Question 3 · part b.iii · 2 marks · Working and answer

Working and answer

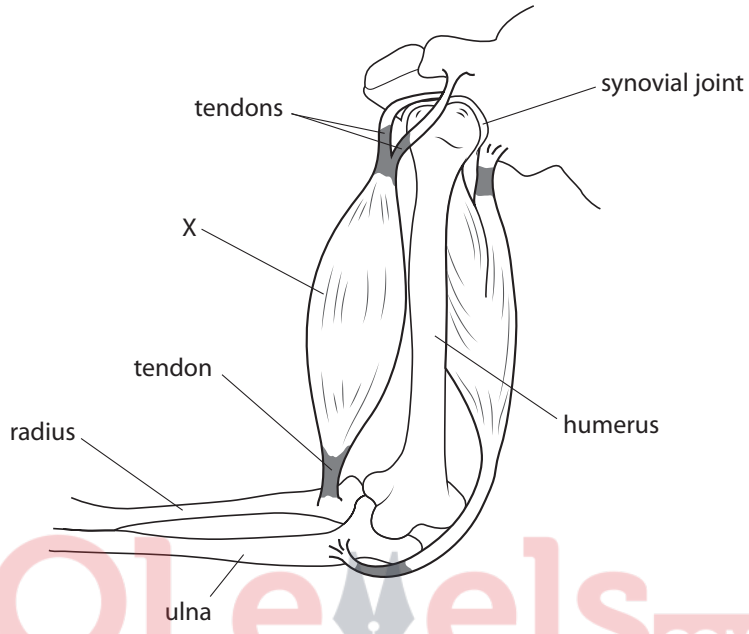
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# 3. Question 7 · part a.i · 1 mark

November 2020 Paper 1

Further practice · Attempt part a.i only.

**7** The diagram shows the bones of a human arm and the muscles that move the lower arm.




(a) (i) Muscle X needs energy to contract.

Name the process that supplies this energy.

**(1)**

.....

(ii) Muscle X contracts when the lower arm is raised further.

Describe the changes in shape of muscle X when the lower arm is raised further.

**(2)**

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**20**

P 6 2 0 5 1 A 0 2 0 2 4

## 3. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

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## 4. Question 5 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

**5** Some athletes train at high altitudes to improve their performance.

Performance improves if more energy is released from aerobic respiration.

(a) (i) Complete the balanced symbol equation for aerobic respiration.

**(2)**

..... + $6O_2$ → ..... + $6CO_2$

(ii) Aerobic respiration produces ATP.

Describe how energy is released from ATP during exercise.

**(2)**

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# 4. Question 5 · part a.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

(b) The table shows the percentage change in mass of haemoglobin in the blood after different periods of time training at altitude.

| Period of time in hours | Percentage change in mass of haemoglobin |
|---|---|
| 0 | 0 |
| 100 | 0.7 |
| 200 | 2.4 |
| 300 | 4.1 |
| 400 | 5.9 |
| 500 | 6.5 |
| 600 | 7.2 |

(i) Plot the results on the grid.

**(4)**

(ii) Draw the curve of best fit.

**(1)**

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## 4. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

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## 5. Question 4 · part a.iv · 1 mark

June 2023 Paper 2

Further practice · Attempt part a.iv only.

**4** (a) (i) Complete the word equation for one type of respiration.

**(2)**

oxygen + ..... → ..... + .....

(ii) State the type of respiration shown by the equation.

**(1)**

.....

(iii) In which part of the cell does this type of respiration occur?

**(1)**

- ☒ **A** endoplasmic reticulum
- ☒ **B** mitochondrion
- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

**(1)**

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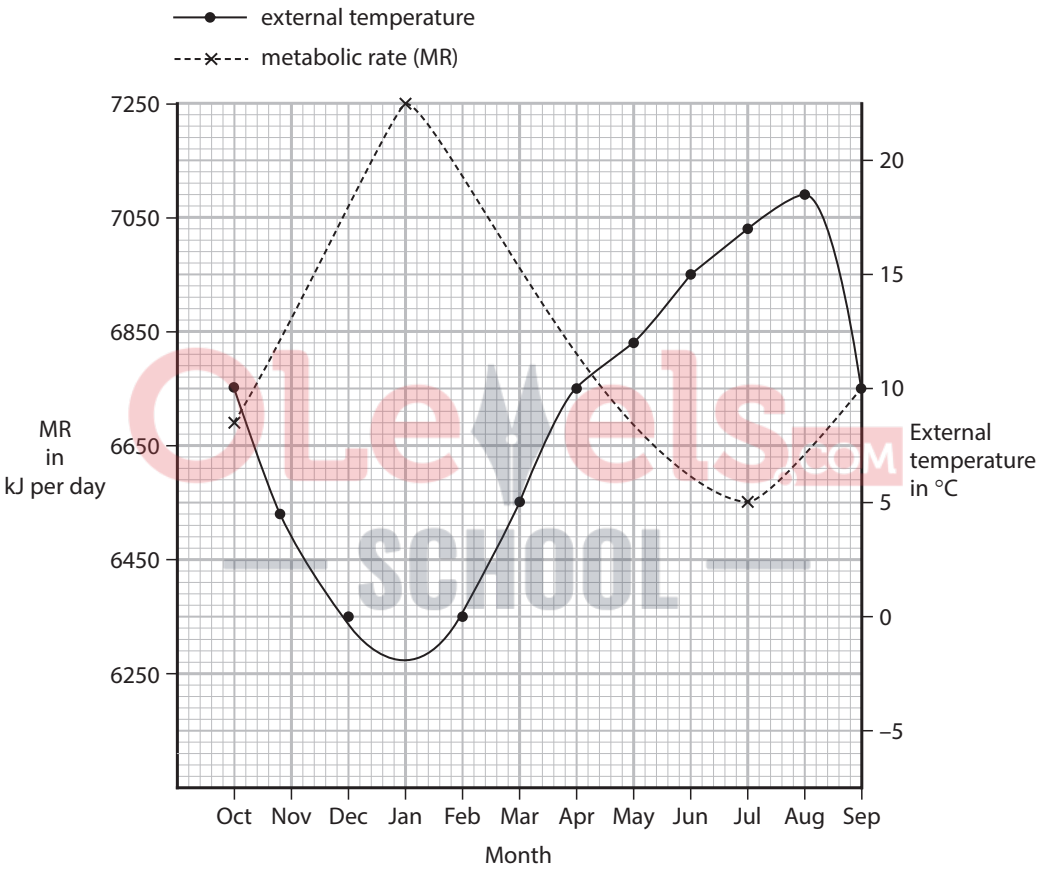
# 5. Question 4 · part a.iv · 1 mark

June 2023 Paper 2

Further practice · Attempt part a.iv only.

(b) An investigation was carried out to see how a person's metabolic rate (MR) and the external temperature change at different times in one year.

The graph shows the results of the investigation for one person.




(i) Describe how the investigation could be made more reliable.

**(2)**

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## 5. Question 4 · part a.iv · 1 mark · Working and answer

Working and answer

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# 6. Question 4 · part b.iii · 5 marks

June 2023 Paper 2

Further practice · Attempt part b.iii only.

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**(2)**

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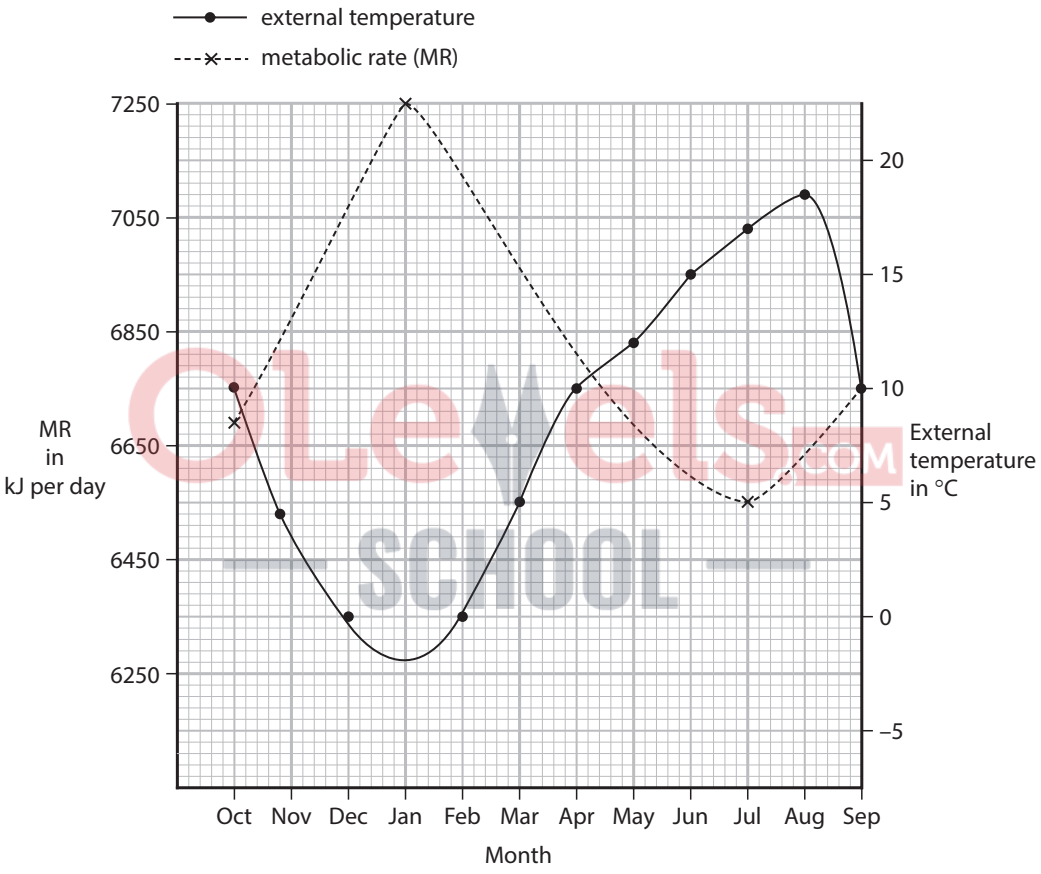
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June 2023 Paper 2

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**(2)**

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# 6. Question 4 · part b.iii · 5 marks

June 2023 Paper 2

Further practice · Attempt part b.iii only.

(ii) Determine the difference between the maximum metabolic rate and minimum metabolic rate.

**(2)**

difference = ..... kJ per day

(iii) Explain why the metabolic rate changes throughout the year.

**(5)**

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## 6. Question 4 · part b.iii · 5 marks

June 2023 Paper 2

Further practice · Attempt part b.iii only.

(iv) Suggest how the metabolic rate could be measured.

**(1)**

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**(Total for Question 4 = 15 marks)**

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Working and answer

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# 7. Question 4 · part b.iv · 1 mark

June 2023 Paper 2

Further practice · Attempt part b.iv only.

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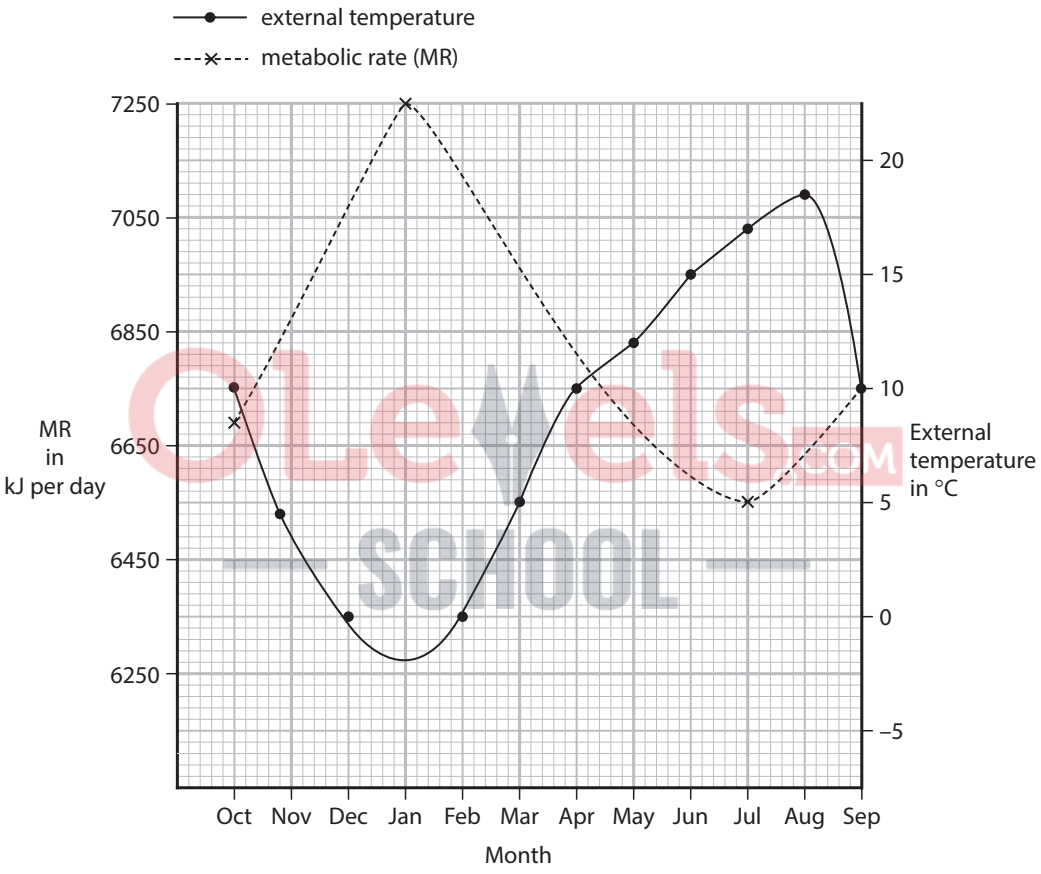
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June 2023 Paper 2

Further practice · Attempt part b.iv only.

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**(5)**

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# 7. Question 4 · part b.iv · 1 mark

June 2023 Paper 2

Further practice · Attempt part b.iv only.

(iv) Suggest how the metabolic rate could be measured.

**(1)**

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**(Total for Question 4 = 15 marks)**

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Working and answer

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